

CSE 4631

Digital Signal Processing

Week 10

Analysis, Calculating the DFT

First Method: Simultaneous equations

- Useful for understanding DFT, but too inefficient for practical use. Requires tremendous number of calculations, and is virtually never used in DSP.
- To calculate N unknown values of the frequency domain from the N values from the time domain, you need N linearly independent equations.
- Summation of the first sample from each sinusoid must be equal to the first sample in the time domain signal equation 1. Likewise, an equation can be written for the remaining points in the time domain signal, resulting in the required N equations.
- Solve them using established methods such as Gauss Elimination.
- However, this shows -
 - Why it is possible to decompose a signal into sinusoids
 - How many sinusoids are needed
 - The basis functions must be linearly independent.

Analysis, Calculating the DFT (contd.)

Second Method: DFT by Correlation

- Standard way of calculating the DFT
- Correlation of the input signal with each basis function —→ ***Analysis equation***

$$ReX[k] = \sum_{i=0}^{N-1} x[i] \cos(2\pi k i / N)$$

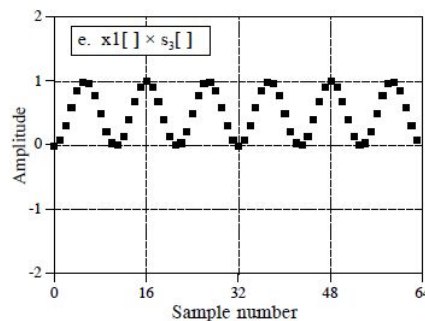
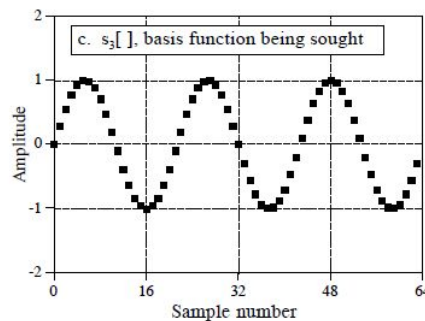
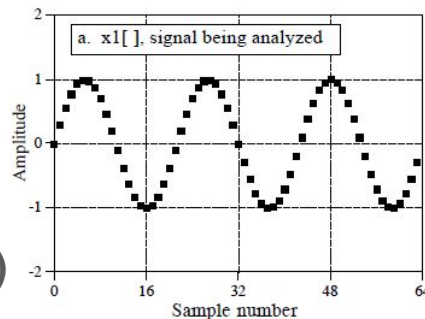
$$ImX[k] = - \sum_{i=0}^{N-1} x[i] \sin(2\pi k i / N)$$

- Each frequency domain sample is found by multiplying the time domain signal by the sine or cosine wave (basis functions).

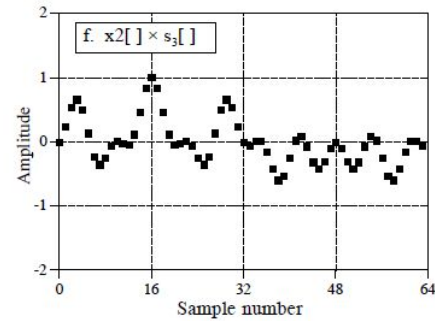
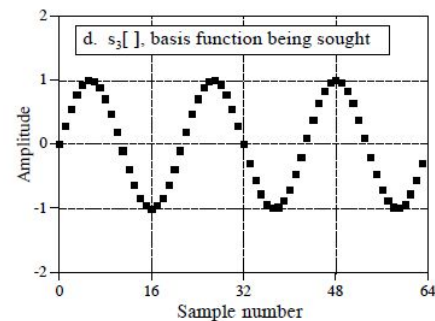
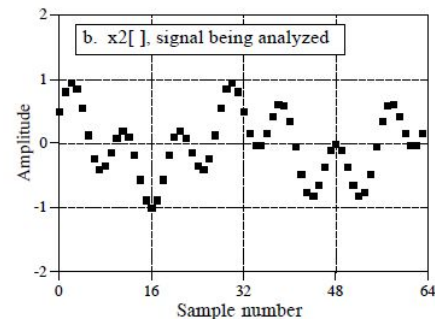
Analysis, Calculating the DFT (contd.)

Second Method: DFT by Correlation (contd.)

Example 1



Example 2



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100 'THE DISCRETE FOURIER TRANSFORM
110 'The frequency domain signals, held in REX[ ] and IMX[ ], are calculated from
120 'the time domain signal, held in XX[ ].
130 '
140 DIM XX[511]           'XX[ ] holds the time domain signal
150 DIM REX[256]          'REX[ ] holds the real part of the frequency domain
160 DIM IMX[256]          'IMX[ ] holds the imaginary part of the frequency domain
170 '
180 PI = 3.14159265       'Set the constant, PI
190 N% = 512              'N% is the number of points in XX[ ]
200 '
210 GOSUB XXXX            'Mythical subroutine to load data into XX[ ]
220 '
230 '
240 FOR K% = 0 TO 256     'Zero REX[ ] & IMX[ ] so they can be used as accumulators
250   REX[K%] = 0
260   IMX[K%] = 0
270 NEXT K%
280 '
290 '                    'Correlate XX[ ] with the cosine and sine waves, Eq. 8-4
300 '
310 FOR K% = 0 TO 256     'K% loops through each sample in REX[ ] and IMX[ ]
320   FOR I% = 0 TO 511   'I% loops through each sample in XX[ ]
330     '
340     REX[K%] = REX[K%] + XX[I%] * COS(2*PI*K%*I%/N%)
350     IMX[K%] = IMX[K%] - XX[I%] * SIN(2*PI*K%*I%/N%)
360     '
370   NEXT I%
380 NEXT K%
390 '
400 END

```

Synthesis, Calculating the Inverse DFT

The **synthesis equation**:

$$x[i] = \sum_{k=0}^{N/2} \text{Re} \bar{X}[k] \cos(2\pi ki / N) + \sum_{k=0}^{N/2} \text{Im} \bar{X}[k] \sin(2\pi ki / N)$$

- $x[i]$ is the signal being synthesized.
- $\text{Re} \bar{X}[k]$ and $\text{Im} \bar{X}[k]$ hold the amplitudes of the cosine and sine waves respectively.
- Index i is running from 0 to $N-1$
- k is running from 0 to $N/2$

Synthesis, Calculating the Inverse DFT (contd.)

The amplitudes needed for synthesis are slightly different from the frequency domain of a signal. In equation form, the conversion between the two is given by:

Equations: Conversion between the sinusoidal amplitudes and the frequency domain values. In these equations, $Re \bar{X}[k]$ and $Im \bar{X}[k]$ hold the amplitudes of the cosine and sine waves needed for synthesis, while $Re X[k]$ and $Im X[k]$ hold the real and imaginary parts of the frequency domain.

$$Re \bar{X}[k] = \frac{Re X[k]}{N/2}$$

$$Im \bar{X}[k] = - \frac{Im X[k]}{N/2}$$

except for two special cases:

$$Re \bar{X}[0] = \frac{Re X[0]}{N}$$

$$Re \bar{X}[N/2] = \frac{Re X[N/2]}{N}$$

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100 'THE INVERSE DISCRETE FOURIER TRANSFORM
110 'The time domain signal, held in XX[ ], is calculated from the frequency domain signals,
120 'held in REX[ ] and IMX[ ].
130 '
140 DIM XX[511]           'XX[ ] holds the time domain signal
150 DIM REX[256]          'REX[ ] holds the real part of the frequency domain
160 DIM IMX[256]          'IMX[ ] holds the imaginary part of the frequency domain
170 '
180 PI = 3.14159265       'Set the constant, PI
190 N% = 512              'N% is the number of points in XX[ ]
200 '
210 GOSUB XXXX             'Mythical subroutine to load data into REX[ ] and IMX[ ]
220 '
230
240 '                     'Find the cosine and sine wave amplitudes using Eq. 8-3
250 FOR K% = 0 TO 256
260   REX[K%] = REX[K%] / (N%/2)
270   IMX[K%] = -IMX[K%] / (N%/2)
280 NEXT K%
290 '
300 REX[0] = REX[0] / 2
310 REX[256] = REX[256] / 2
320 '
330 '
340 FOR I% = 0 TO 511     'Zero XX[ ] so it can be used as an accumulator
350   XX[I%] = 0
360 NEXT I%
370 '

```


380 '	Eq. 8-2 SYNTHESIS METHOD #2. Loop through each
390 '	sample in the time domain, and sum the corresponding
400 '	samples from each cosine and sine wave
410 '	
420 FOR I% = 0 TO 511	'I% loops through each sample in XX[]
430 FOR K% = 0 TO 256	'K% loops through each sample in REX[] and IMX[]
440 '	
450 XX[I%] = XX[I%] + REX[K%] * COS(2*PI*K%*I%/N%)	
460 XX[I%] = XX[I%] + IMX[K%] * SIN(2*PI*K%*I%/N%)	
470 '	
480 NEXT K%	
490 NEXT I%	
500 '	
510 END	

Duality

- *DFT* and *Inverse DFT* use the same mathematical approach: known values are multiplied by the basis functions, and the resulting products are added.
- Only significant difference is: Time domain has one signal of N points and the frequency domain has two signal of $N/2+1$ points
- This symmetry between the time and frequency domains is called **duality**.
- Important properties arising from duality:
 - A single point in the time domain corresponds to a sinusoid in the frequency domain. And the opposite is also true
 - Convolution in the time domain corresponds to multiplication in the frequency domain. And the reverse is also true.

Polar Notation

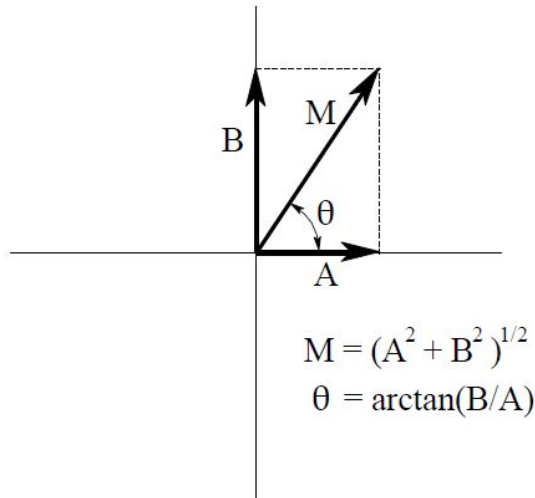
- The frequency domain is a group of amplitudes of cosine and sine waves (with slight scaling modifications). This is called **rectangular notation**.
- Alternatively, the frequency domain can be expressed in **polar** form.
- In this notation, $Re\ X[]$ & $Im\ X[]$ are replaced with two other arrays
- **Magnitude of $X[]$** , written in equations as: $Mag\ X[]$, and the **Phase of $X[]$** , written as: $Phase\ X[]$.
- The magnitude and phase are a pair-for-pair replacement for the real and imaginary parts.
- $Mag\ X[0]$ and $Phase\ X[0]$ are calculated using only $Re\ X[0]$ and $Im\ X[0]$
- Adding a cosine wave and a sine wave of the same frequency results in a cosine wave of the same frequency, but with a new amplitude and a new phase shift. In equation:

$$A \cos(x) + B \sin(x) = M \cos(x + \theta)$$

Polar Notation (contd.)

Conversion between notations:

- Follows the same conversion equations as do simple vectors.
- Consider this vector representation of how the two variables A and B, can be viewed in a rectangular coordinate system, while M and θ are parameters in polar coordinates



$$Mag X[k] = \sqrt{ReX[k]^2 + ImX[k]^2}$$

$$Phase X[k] = \tan^{-1} \left(\frac{ImX[k]}{ReX[k]} \right)$$

$$ReX[k] = MagX[k] \cos(PhaseX[k])$$

$$ImX[k] = MagX[k] \sin(PhaseX[k])$$

Polar Notation (contd.)

- Use case
 - Rectangular notation is usually the best choice for calculations, such as in equations and computer programs
 - Polar form is used to form graphs
- Why is it easier to understand the frequency domain in polar notation?
 - Polar notation directly represents signals in terms of the amplitude and phase of the component cosine waves.
 - In turn, systems can be represented by how they modify the amplitude and phase of each of these cosine waves

Polar Nuisances

Nuisance 1: Radians vs Degrees

- When expressed in degrees, the values in the phase signal are between -180 and 180
- Using radians, each of these values will be between $-\pi$ and π , that is, between -3.141592 to 3.141592.
- Most computer languages require the use of radians for their trigonometric functions, such as cosine, sine, arctan, etc.
- It can be irritating to work with long decimals and difficult to interpret the received data
- **Solution:** Define the constant, $PI = 3.141592$ at the beginning of your program

Polar Nuisances (contd.)

Nuisance 2: Divide by zero error

- Real part is zero and the imaginary part is some nonzero value.
- This means the phase is exactly 90 or -90 degrees.
- **Solution**
 - Test if the real part is zero, and if it is, test if the imaginary part is positive or negative, to determine whether to set the phase to $\pi/2$ or $-\pi/2$ respectively. Then bypass the division.
 - Test if the real part is zero, and if it is, change it to a negligibly small number.

Polar Nuisances (contd.)

Nuisance 3: Incorrect arctan

- If $Re X[k] = 1$ and $Im X[k] = 1$, then the corresponding polar values are $Mag X[k] = 1.414$ and $Phase X[k] = 45^\circ$.
- If $Re X[k] = -1$ and $Im X[k] = -1$, then the corresponding polar values are also $Mag X[k] = 1.414$ and $Phase X[k] = 45^\circ$. Hence, the phase is wrong!
- Occurs when the real part is negative.
- **Solution (Test if the real and imaginary parts are negative)**
 - When both of them are negative, subtract 180° (π radians) from the calculated phase.
 - If only the real part is negative and the imaginary part is positive, then add 180° .

Polar Nuisances (contd.)

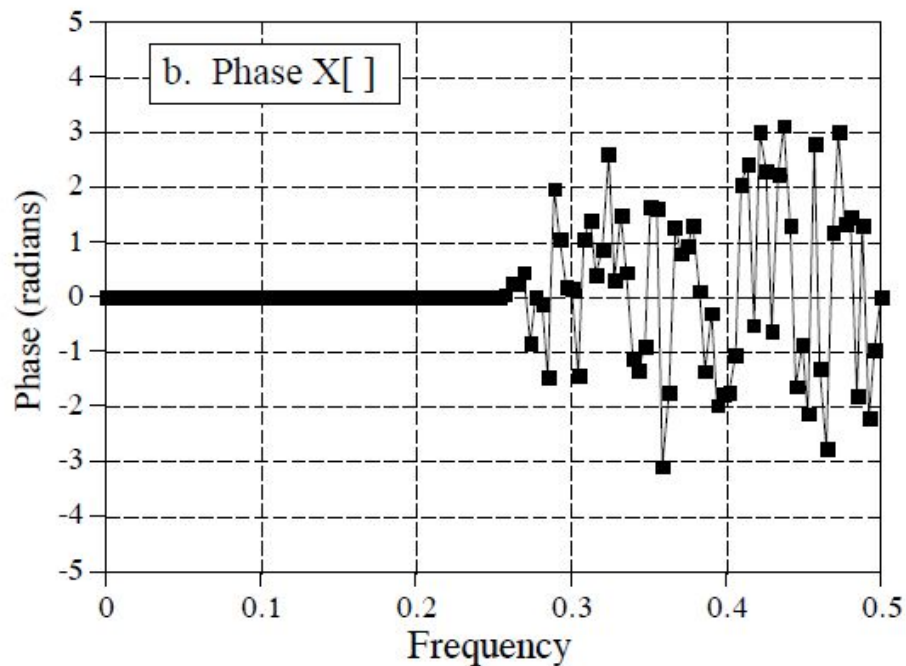
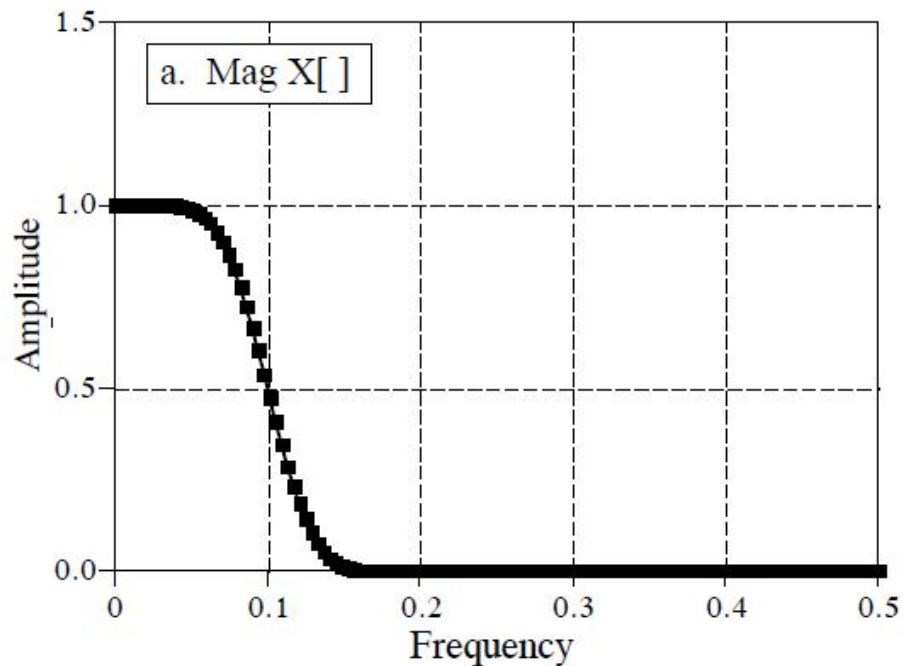
Nuisance 4: Phase of very small magnitudes

- If the magnitude is negligibly small, the phase doesn't have any meaning and can assume unusual values
- It means that the corresponding values in the amplitude signal are so small that the signal is buried and lost in round-off noise.
- The values in the phase, when a polar signal is contaminated with noise, are random numbers between $-\pi$ and π . It might often look like a real signal, but they are not.

Note: Round-off noise refers to the error introduced when real-valued numbers are approximated due to the limited precision of digital hardware

Polar Nuisances (contd.)

Nuisance 4: Phase of very small magnitudes (contd.)



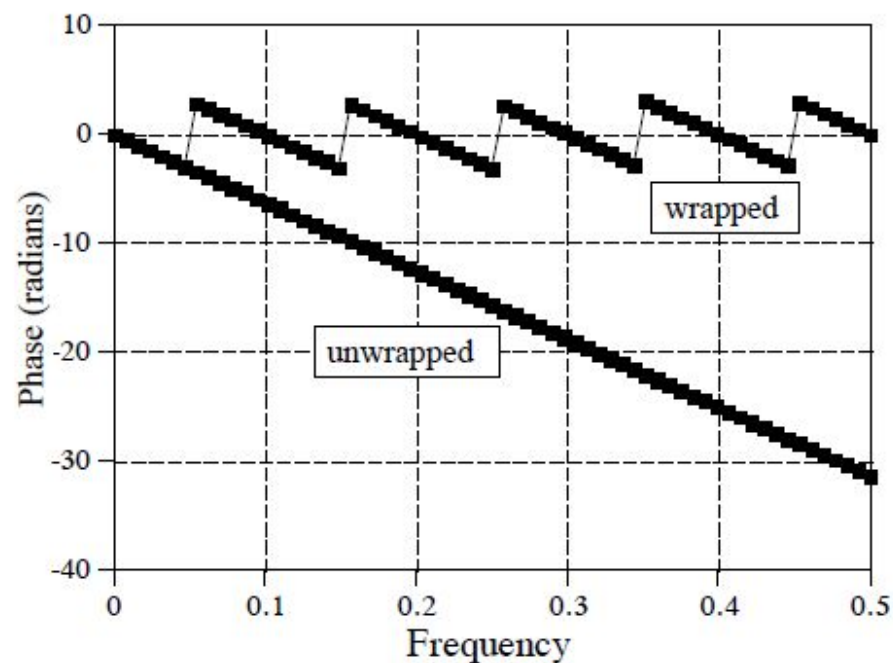
Polar Nuisances (contd.)

Nuisance 5: 2π ambiguity of the phase

- A phase shift of θ is exactly the same as a phase shift of $\theta+2\pi$, $\theta+4\pi$, $\theta+6\pi$, etc.
- Any sinusoid is unchanged when you add an integer multiple of 2π to that phase.
- This creates several discontinuities in the data. The phase snaps back to 3.141592 as soon as it is going to dip below -3.141592.
- **Why?** Computer algorithm picks its favorite choice from an infinite number of equivalent possibilities – the smallest possible value – keeping the phase between $-\pi$ and π
- **Solution:** It is often easier to understand the phase if it does not have these discontinuities, even if it means that the phase extends above π , or below $-\pi$. This is called **unwrapping the phase**.
 - A multiple of 2π is added or subtracted from each value of the phase.

FIGURE 8-12

Example of phase unwrapping. The top curve shows a typical phase signal obtained from a rectangular-to-polar conversion routine. Each value in the signal must be between $-\pi$ and π (i.e., -3.14159 and 3.14159). As shown in the lower curve, the phase can be *unwrapped* by adding or subtracting integer multiples of 2π from each sample, where the integer is chosen to minimize the discontinuities between points.



```

100 ' PHASE UNWRAPPING
110 '
120 DIM PHASE[256]           'PHASE[ ] holds the original phase
130 DIM UWPHASE[256]        'UWPHASE[ ] holds the unwrapped phase
140 '
150 PI = 3.14159265
160 '
170 GOSUB XXXX               'Mythical subroutine to load data into PHASE[ ]
180 '
190 UWPHASE[0] = 0           'The first point of all phase signals is zero
200 '
210 '                       'Go through the unwrapping algorithm
220 FOR K% = 1 TO 256
230  C% = CINT( (UWPHASE[K%-1] - PHASE[K%]) / (2 * PI) )
240  UWPHASE[K%] = PHASE[K%] + C%*2*PI
250 NEXT K%
260 '
270 END

```

Polar Nuisances (contd.)

Nuisance 6: The magnitude is always positive (π ambiguity of the phase)

- If the imaginary part in rectangular notation is *entirely zero*, the polar signals contain discontinuities and sharp corners.
- This is because the magnitude must always be positive, *by definition*. So, whenever the real part dips below zero, the magnitude remains positive by changing the phase by π (or $-\pi$).
- This is not a problem for the mathematics, but the irregular curves can be difficult to interpret.

$$Mag\ X[k] = \sqrt{ReX[k]^2 + ImX[k]^2}$$

$$Phase\ X[k] = \tan^{-1}\left(\frac{ImX[k]}{ReX[k]}\right)$$

$$ReX[k] = MagX[k] \cos(PhaseX[k])$$

$$ImX[k] = MagX[k] \sin(PhaseX[k])$$

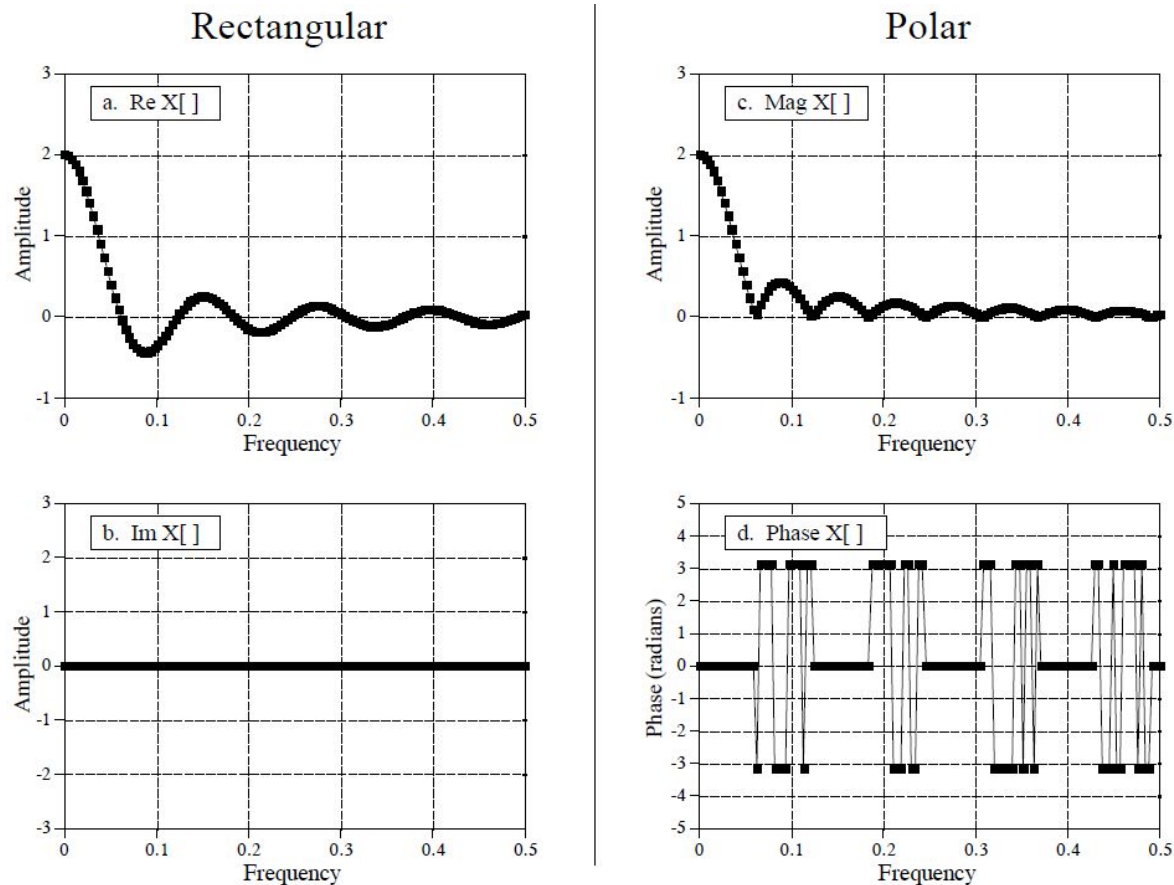


FIGURE 8-13

Example signals in rectangular and polar form. Since the magnitude must always be positive (by definition), the magnitude and phase may contain abrupt discontinuities and sharp corners. Figure (d) also shows another nuisance: random noise can cause the phase to rapidly oscillate between π or $-\pi$.

Polar Nuisances (contd.)

Nuisance 7: Spikes between π and $-\pi$

- Since π and $-\pi$ represent the same phase shift, round-off noise can cause adjacent points in the phase to rapidly switch between the two values
- This can produce sharp breaks and spikes in an otherwise smooth curve.
- But the phase isn't really this discontinuous! It is only the noise.